

From System Ranking to Decision Value: How Much Human Evaluation Do Automatic Evaluators Actually Save?

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Abstract

Automatic evaluators, from learned metrics to LLM judges, are increasingly credited with the human labels they save, and that credit is usually inferred from their correlation with human scores. We measure it directly for a concrete decision: certifying, from a human audit, that a selected system is within ε of the best of a menu of close competitors. An evaluator’s human-evidence saving (HES) is the fraction of human labels, pilot included, that it saves for this decision over the best fixed annotation design that uses no evaluator. Across machine translation (WMT22 MQM; 31 segment-level metric variants and open LLM judges), chat (LMArena) and retrieval, with pre-registered MT and chat tests, evaluator quality turns out to depend on the level at which it is measured: metrics with system-level correlations of 0.93–0.99 reach correlations of only 0.11–0.38 on the paired differences a close decision depends on. Correspondingly, over 33 informative cells from 19 two-system MT decisions (300 simulated audits, 34 evaluators), the mean evaluator saves a median 0–2% of human labels on top of a judge-free design, which itself saves a median 5.5% (8–16% in the four pre-registered cells, mostly by not re-rating identical outputs; 41–56% in retrieval). With 2,000 audits per cell (31 metric variants) the picture sharpens rather than changes: the design saves 4.6%, the mean evaluator 2.3%, the ten strongest 3.9% and the post-hoc best 6.0% with a refitted coefficient, and a 10% saving is ruled out for 94% of decision–evaluator pairs. Realised savings follow $\rho^2(1 - P/J)$, so they are a property of the regime as much as of the evaluator: with decision-level $\rho \leq 0.43$ and a pilot that takes up to two thirds of the cost, this ceiling is below 5% in 94% of cases, and even a negligible pilot would leave it at $\rho^2 \leq 0.19$. Freezing the correction coefficient on a 50-item pilot costs a further 2.5–4.3 points, which refitting on all labels removes in our simulations, with average coverage near nominal.

1 Introduction

Human judgments remain the reference for comparing NLP systems, and they are expensive. Automatic evaluators promise to stretch them: score everything automatically and spend human effort only on correcting the evaluator’s bias, as in prediction-powered inference (PPI; Angelopoulos et al., 2023a,b). How much an evaluator saves under such a correction is now measured directly (Boyeau et al., 2024; Chen et al., 2026), and Gao et al. (2026) propose to rank MT metrics by the annotation they save (PPSR), which is governed by the squared correlation between human and metric score *differences*.

PPSR is already a realised quantity: Gao et al. (2026) verify by resampling that, under uniform labelling with the coefficient fitted on the evaluation labels, the strongest WMT22 metrics save about 9–15% of the labels a paired comparison needs. We do not revisit that number. We ask how much of it is left for the decision an evaluation is usually run for, and how much of it a competent annotation design obtains without any evaluator. A team has a short list of strong systems that are close in quality, and wants to certify from a human audit that the one it picks is within ε of the best. Three things separate this setting from the asymptotic picture. First, the decision is local: the systems that matter are the top few, not the whole field over which system-level correlation is computed. Second, a human audit need not sample uniformly: items whose paired difference is known to be zero (identical outputs, documents outside every system’s cutoff) need no label, and others can be sampled in proportion to how much they can move the decision; these labels are saved without any evaluator, which should be credited only with what it adds on top. Third, the coefficient that corrects the evaluator must itself be estimated from the few human labels the audit has, or be fitted on all of them with a variance estimate

that ignores the coefficient’s error.

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Contributions.

1. **A decision-level measure of evaluator value that credits the design first.** The human-evidence saving $HES = 1 - J_{50}(\text{design} + \text{evaluator}) / J_{50}(\text{best fixed judge-free design})$, with J_{50} the number of unique human labels, pilot included, at which the decision is certified with probability $1/2$ (§2).
2. **The design is credited first and usually saves more; the evaluator’s increment is small on average and depends on the evaluator.** Over 33 informative cells from 19 two-system WMT22 decisions (300 audits, 34 evaluators), a judge-free design saves a median 5.5% of uniform-sampling labels; the mean evaluator adds 0–2% on the same design (0.6–2.8% for the ten with the highest system-level correlation) and the design saves more in 76–94% of cells under every coefficient rule. In the four pre-registered cells the best of 31 metric variants adds at most 7% to a design saving of 8–16% with the pilot-fixed coefficient (refitted: up to 15%). Retrieval: 41–56% vs. 1–9 further points (§5).
3. **What the decisions rule out.** With 2,000 simulated audits per cell (31 metric variants) the mean evaluator saves 2.3%, the ten strongest 3.9% and the post-hoc best 6.0% (refitted), and a one-sided 95% bound on a single evaluator’s saving over the design is below 5% in 64% and below 10% in 94% of decision–evaluator pairs (29% and 88% for the ten strongest): the savings are real, small on average, and bounded (§5).
4. **Evaluator quality collapses across levels.** Metric variants with system-level Pearson 0.93–0.99 reach decision-level correlations of only 0.11–0.38; 10 of the 31 rank the top-4 zh→en systems in exactly reversed order (§4).
5. **Savings are capped by decision-level ρ and the pilot.** With the coefficient refitted on all labels, realised HES follows the tax-free ceiling $\rho^2(1 - P/J)$, which is below 5% in 94% of decision–evaluator pairs. Freezing the coefficient on a 50-item pilot costs a further 4.3 [3.5, 5.2] (en→de) and 2.5 [2.0, 3.1] (zh→en) points over the 31 variants; refitting removes

this tax with nominal coverage in our simulations, and cross-fitting only partly. Smaller pilots (25, 10 segments) do not change the picture: the mean evaluator then adds 0.3–0.7% refitted and loses 3–8% with a frozen coefficient (§6).

6. **Pre-registered tests, with failures reported.** Three locks fixed hypotheses before the corresponding evaluator outputs were joined with human labels (limits in §3); of ten hypotheses, H2, H5 and H6 failed and H4 failed in chat (§8).

We propose no estimator: the corrected estimator is PPI++ with a pilot-fixed coefficient, and the allocation is active statistical inference (Zrnica and Candès, 2024a) for the decision’s functional.

2 Setup

Decision. A menu $\mathcal{M}$ of systems is evaluated on a fixed set of N items with human utilities $u_m(i)$ and means μ_m . An audit labels a pilot of P items fully, picks the candidate $\hat{m}$ with the best pilot mean, and *certifies* it if for every competitor j a one-sided upper bound on $\mu_j - \mu_{\hat{m}}$ is at most ε (normal bound, Bonferroni over the $|\mathcal{M}| - 1$ competitors, $\alpha = 0.10$). A certificate is wrong if $\max_j \mu_j - \mu_{\hat{m}} > \varepsilon$. Menus are deliberately hard, built from the full human labels (the top-4 systems by MQM; the closest Arena pairs); their construction is not part of the audit.

Estimator. After the pilot, items are drawn by Poisson sampling with probabilities π_i . For $D_j = u_j - u_{\hat{m}}$ and an evaluator’s predicted difference $\hat{D}_j$,

$$\hat{\Delta}_j = \frac{1}{N} \left[\sum_{\text{pilot}} D_j + \sum_{\text{rest}} \lambda_j \hat{D}_j + \sum_{i \in S} \frac{D_j(i) - \lambda_j \hat{D}_j(i)}{\pi_i} \right], \quad (1)$$

with $\lambda_j = \max(0, \widehat{\text{Cov}}(D_j, \hat{D}_j) / \widehat{\text{Var}}(\hat{D}_j))$ fitted on the pilot and frozen; $\lambda = 0$ is the human-only estimator. Given the pilot, (1) is unbiased for the finite-population mean for any evaluator; its variance is estimated by $\sum_S (1 - \pi_i) R_i^2 / (\pi_i^2 N^2)$, $R = D - \lambda \hat{D}$.

Designs. *Uniform:* constant π ; every menu output of a sampled item is labelled, identical strings included, as in WMT22’s own MQM campaign, which rated each system’s output separately (42% of en→de and 18% of zh→en rated outputs share

their string with another system’s; 26% of such groups received different scores). We treat those differences as rater noise around one quality per string: the estimand averages the ratings of identical strings, whose paired difference is then zero; sharing one random rating instead changes little (App. C). *Dedup*: uniform over items where some competitor’s output differs from the candidate’s, identical strings rated once, and $D_j = 0$ used exactly where outputs coincide. *Weighted*: as dedup, with items sampled in proportion to a label-free proxy of $|D|$ (in retrieval the closed-form decision weight of each document; in MT $\sqrt{\sum_j \hat{\sigma}_j^2(b)}$, with $\hat{\sigma}_j^2(b)$ the pilot mean of D_j^2 in one of three bins b of the chrF-based dissimilarity of the two outputs: below its median, its third and its fourth quartile; identical outputs form a fourth bin of weight zero. A four-quartile variant changes nothing of substance (App. C). The *best fixed judge-free baseline* of a cell is the cheapest of these human-only designs there; it is a benchmark, not a procedure an auditor selects. An evaluator enters as the control variate of (1) on top of a design.

Cost. J_{50} counts unique human labels, pilot included, at which a pre-committed budget certifies in half of 300 simulated audits (linear interpolation over a geometric grid of 14 budgets up to the whole post-pilot set). The 50-segment MT pilot costs 162 (en→de) and 182 (zh→en) labels, 42–68% of J_{50} on en→de; an evaluator can act only on the rest (the factor $1 - P/J$ of §6). Intervals are paired bootstraps over the 300 simulated audits of a fixed test set: they reflect Monte-Carlo, not data-sampling, uncertainty.

3 Data, evaluators and protocol

MT: WMT22 MQM (Freitag et al., 2021, 2022) en→de (1,315 segments) and zh→en (1,875), Google weights, $u = -\min(\text{MQM}, 25)/25$, identical output strings of a segment share the mean of their ratings; menus of the four best WMT submissions (MBR systems excluded). Evaluators: the 31 WMT22 metric variants available at segment level in mt-metrics-eval (Freitag et al., 2022) (reference-based variants scored against refA, including COMET-22 (Rei et al., 2022), and reference-free QE variants), plus our own COMET-22 run and Qwen3-8B (Yang et al., 2025) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) with the GEMBA-DA prompt (Kocmi and Federmann,

2023). **Chat:** LMArena human-preference battles (Chiang et al., 2024); the six English single-turn pairs with the smallest win-rate gaps among those with ≥ 300 battles (311–472 battles; gaps 0.011–0.084). Evaluators: Qwen3-8B and Mistral-7B as pairwise judges (both presentation orders averaged), “the longer response wins”. **Retrieval:** a re-analysis of fixed-budget audits from an earlier study (Anonymous, 2026): three cutoff policies over a shared ranked pool (an expected-F1 gate, a score threshold, a learned truncation) compared by precision at their cutoffs, ε in precision units, on four fully judged collections (ANTIQUE, CAsT 2019, DBpedia-Entity, TREC DL 21–23; Hashemi et al., 2020; Dalton et al., 2020; Hasibi et al., 2017; Craswell et al., 2021, 2022, 2023) with Qwen3-8B (UMBRELA prompt; Upadhyay et al., 2024) and Qwen3-Reranker-0.6B (Zhang et al., 2025) as judges; pilot 20 queries. MT and chat use a 50-item pilot (ablations: 100 for MT, 25 for chat), $\varepsilon \in \{0.01, 0.02\}$ (MT) and $\{0.05, 0.1\}$ (chat).

Pre-registration. Lock v0.8 (MT, chat) was committed before any LLM or COMET judgment was joined with a human label; v0.9 replaced the uninformative v0.8 chat decision (gaps 0.10–0.56, settled by the pilot alone) by close pairs before they were judged; v1.1 fixed the 31-metric test before any of its savings were computed. All locks were committed within about four hours on public data whose human labels exist; before v0.8 the human-only MT designs had been run in development (they saved 8–19% over uniform sampling and chrF about nothing, so H1 was not a blind test), and v1.1 followed the exploratory estimation-tax analyses of the open judges, so H9 tests on 31 new evaluators a phenomenon first seen on four. All locks, deviations and outcomes are in the supplementary material.

4 Evaluator quality depends on the level of measurement

Figure 1 places the 30 positively correlated WMT22 metric variants (REUSE omitted) on three levels against their realised HES. At the global level (Pearson over the 11 WMT submissions) nearly all strong metrics are above 0.9. Among the four closest systems the ranking is fragile: MetricX-XL-DA and MetricX-XXL-DA reach system-level Pearson 0.994–0.995 on zh→en and Kendall $\tau = 0$ among the top four. The second- and third-ranked systems differ by only 0.0009 (en→de) and 0.0017

(zh→en) in utility (standard error 0.002–0.004), so $\tau = 0$ can reflect fragility; but 10 of the 31 zh→en variants (system-level Pearson 0.83–0.96; BLEU, spBLEU, chrF, BERTScore, YiSi, SEScore, MEE) reach $\tau = -1$, which also reverses the first-ranked system’s lead over each of the other three (2.3–3.5 standard errors). The quantity that governs a control variate is the correlation ρ of per-item paired differences, and it is low everywhere: 0.30–0.38 for the best metrics on en→de (mean over the menu’s pairs), 0.21–0.25 on zh→en, and 0.17–0.29 for COMET-22 and the Qwen3-8B GEMBA judge (system-level 0.97–0.99; here medians over pairs of the top-6 systems), while the Mistral-7B GEMBA judge reaches only 0.04–0.09. Part of this is label noise: in the en→de three-rater re-annotation, two raters’ paired differences on the same items correlate only 0.31–0.41, so even a perfect evaluator would reach $\rho \approx 0.59$ against the single ratings an audit collects; the best metrics attain one half to two thirds of that (App. C). Across metrics, HES is ranked slightly better by ρ than by system-level Pearson (Spearman 0.39 vs. 0.36 on en→de, 0.22 vs. 0.07 on zh→en; H11), but neither predicts it well: all real metrics sit where the saving is small (§6).

5 Where the savings come from

Figure 3 and Table 1 decompose the saving over uniform sampling into the part a judge-free design obtains and the part the best evaluator adds on top. In retrieval, skipping documents that no comparison depends on and sampling the rest by decision weight saves 41–56% of the labels of uniform sampling; the Qwen3-8B judge adds 1–9 points of that total (HES 3–20% over the design). In MT, identical outputs (per pair of menu systems, 14–24% of en→de and 5–16% of zh→en segments) need no label, and many paired MQM differences are exactly zero (53–59% and 40–48%); exploiting this saves 16% (en→de) and 8% (zh→en) at $\varepsilon = 0.01$. The best of the 31 metric variants, chosen *post hoc* and therefore favoured, adds 7% on the same design in both language pairs, and the median metric adds nothing (–1% to 0%). In all four pre-registered MT cells the design’s saving (8–16%) exceeds the best metric’s increment (1–7%; H8, 4/4; H1 with the open judges, 4/4). Most of this is deduplication: not re-rating identical outputs saves 13% and 6% on en→de and all 8% on zh→en.

Across 30 decisions. Four cells are few, and their Monte-Carlo intervals are wide. We therefore repeated the comparison (exploratory, post-lock) on every two-system decision among the top six WMT22 systems of each language pair: 30 decisions, two ε each, all 31 metric variants, COMET-22 and both GEMBA judges (Figure 2). In 33 of the 60 cells, from 19 decisions (4 en→de, 29 zh→en), at least 30% of the uniform-sampling cost falls after the pilot (App. C); in the others the pilot nearly settles the decision. Over these 33 cells the judge-free design saves a median 5.5% (interquartile range 3–10%), and the mean evaluator a median –0.1% with the pilot coefficient, 1.6% refitted and 0.4% cross-fitted; the design saves more than the mean evaluator in 94%, 82% and 91% of the cells (91%, 76% and 79% for the ten evaluators with the highest system-level correlation). The best evaluator chosen *post hoc* beats the design in about half of them; at 300 audits its lead over the mean evaluator (median 4.3–5.0 points) matches what picking the largest of 34 equal evaluators produces under the Monte-Carlo noise (4.3–4.8 points; significant at 0.05 in at most 6% of cells), so that test is uninformative and is repeated below with 2,000 audits. Only 5.5% of MT decision–evaluator pairs have a refitted HES interval above zero.

What the decisions rule out. With 300 audits the median 95% interval of one evaluator’s HES is ± 6.6 points, too wide to call a saving small. We therefore re-ran the 30 decisions with 2,000 audits per cell (the 31 metric variants, otherwise identical), which narrows the interval to ± 2.3 points; 36 informative cells. The medians hold: the design saves 4.6%, the mean evaluator 0.2 / 2.3 / 0.9% (pilot / refit / cross-fit), and the design saves more in 100 / 89 / 100% of cells (89 / 61 / 78% against the mean of the ten strongest evaluators, which reaches 3.9% refitted). The *post-hoc* best (median 6.0% refitted) now leads the mean by more than selection noise explains (3.3 vs. 1.8 points; $p < 0.05$ in 72% of cells refitted, 33% pilot, 28% cross-fit), so evaluators do differ; the design beats it in 39% of cells refitted and 61% under the other rules. With the refitted coefficient the one-sided 95% upper bound of a single evaluator’s HES over the design is below 5% in 64% and below 10% in 94% of the 1,116 decision–evaluator pairs (29% and 88% for the ten evaluators with the highest system-level correlation; 89% and 99% with the pilot coefficient). For the *mean* evaluator of a cell the bound is be-

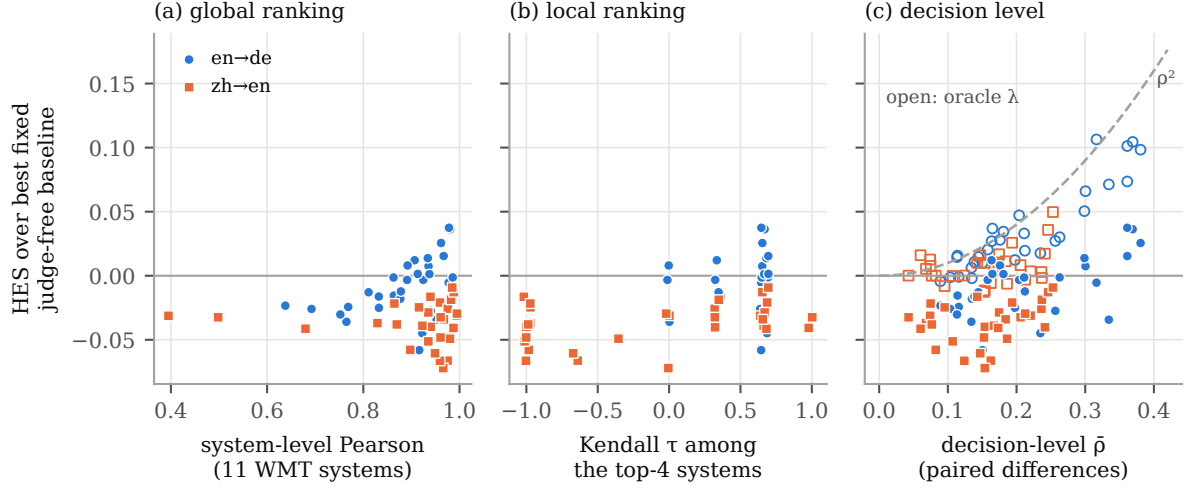


Figure 1: Thirty WMT22 metric variants at three levels of quality against their realised human-evidence saving over the best fixed judge-free baseline (pilot 50, mean over $\varepsilon \in \{0.01, 0.02\}$). (c) open markers: the same metrics with the population-optimal (oracle) correction coefficient; dashed: ρ^2 , the asymptotic saving. REUSE (negative system-level correlation; switched off by $\lambda \geq 0$) is omitted.

low 5% in 89% and below 10% in 97% of cells; the exception is one $\text{en} \rightarrow \text{de}$ decision at $\varepsilon = 0.01$, where the mean evaluator saves 7% (bound 13%) and the design 14%. Many small refitted savings are now distinguishable from zero (53% of pairs; median 2.0%, interquartile range 0.8–3.6%): real, and small. In labels: of a median decision’s 256, the design saves 17, the mean evaluator 6 (refitted), the post-hoc best 17.

In retrieval, the evaluator’s gain across 54 settings of an earlier study (Anonymous, 2026) correlated 0.75 with ρ and 0.03 with label accuracy; on the three collections held out there (NeuCLIR-Bench, TREC-COVID, Touché-2020; Lawrie et al., 2025; Voorhees et al., 2020; Bondarenko et al., 2020) the design saves 6–50% of uniform-sampling labels and the judge adds HES 2–16%; in the locked NeuCLIRBench test a 20-query pilot over-estimated the judge’s increment (cost ratio 0.71–0.76 predicted, 0.82–0.93 realised).

6 From correlation potential to realised saving

Proposition 1 (Finite-pilot cost approximation). *Let a design need L_0 post-pilot labels for a fixed decision without an evaluator, and let the evaluator’s paired differences $\hat{D}$ have correlation ρ with the human ones D . Assume that, over informative items (those with a non-zero difference), D is a linear projection on $\hat{D}$ with a residual whose variance does not depend on $\hat{D}$ (homoskedastic;*

Cell	378	uniform J_{50}	judge-free best fixed	+ evaluator best (median)	evaluator HES
IR ANTIQUE	379	1,405	836	770	8%
IR CAsT 2019		4,778	2,127	1,720	19%
IR DBpedia-Entity		5,563	2,543	2,046	20%
IR DL 21–23		4,092	1,783	1,722	3%
MT $\text{en} \rightarrow \text{de}$, $\varepsilon=0.01$		463	389 (w)	362 (390)	7% (0%)
MT $\text{en} \rightarrow \text{de}$, $\varepsilon=0.02$		261	238 (w)	233 (242)	2% (–2%)
MT $\text{zh} \rightarrow \text{en}$, $\varepsilon=0.01$		2,990	2,738 (d)	2,539 (2,760)	7% (–1%)
MT $\text{zh} \rightarrow \text{en}$, $\varepsilon=0.02$		1,166	1,066 (d)	1,052 (1,104)	1% (–4%)

Table 1: Unique human labels (J_{50} , pilot included) for uniform sampling, the best fixed judge-free design, and the *same* design with an evaluator added as control variate. Retrieval: Qwen3-8B judge, $\varepsilon = 0.02$, pilot 20 queries. MT: the best of the 31 WMT22 metric variants chosen post hoc, and the median metric in parentheses, on the cell’s best fixed design (w: weighted, d: dedup); pilot 50 segments, pilot-fixed coefficient. 95% Monte-Carlo intervals: retrieval HES [5, 10], [14, 24], [13, 27] and [13, 17]%; MT design savings [10, 29], [–13, 15], [3, 14] and [6, 15]%, best-metric HES [–7, 12], [–15, 6], [–6, 19] and [–3, 8]%; single MT cells are weak evidence, and Figure 2 pools 33. H8 as locked used the weighted design for every metric (App. C).

joint normality is a sufficient case), and that λ is fitted by least squares on a pilot independent of the post-pilot sample, with effective informative size P_{eff} (the number of pilot items that contribute variation to the paired regression, approximated by those with a non-zero paired difference; λ_j and P_{eff} are per competitor j). Then, to first order in $1/P_{\text{eff}}$ and under the normal fixed-budget approximation, the variance of the corrected estimator is

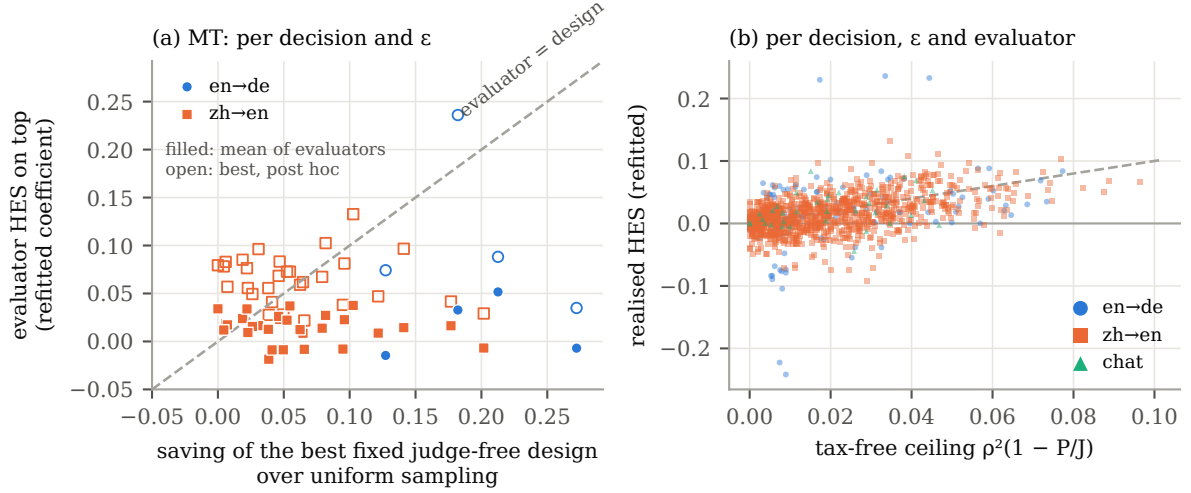


Figure 2: The 30 two-system WMT22 decisions (exploratory). (a) The 33 MT decision- ϵ cells (19 decisions) in which at least 30% of the uniform-sampling cost falls after the pilot: saving of the best fixed judge-free design over uniform sampling against the HES of the evaluators on top of it (refitted coefficient; filled: mean of the 34 evaluators, open: best, post hoc). (b) Every informative decision, ϵ and evaluator, MT and chat (1,158 points): realised HES against the tax-free ceiling $\rho^2(1 - P/J)$ of (2), with the pilot-median ρ .

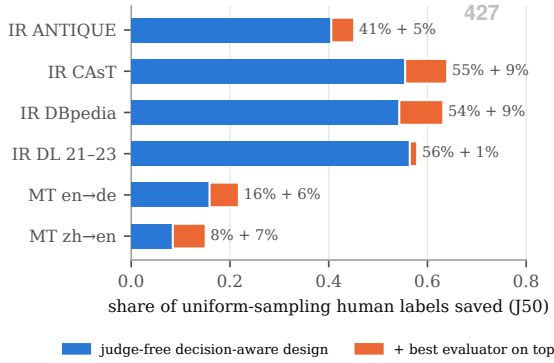


Figure 3: Share of uniform-sampling human labels (J_{50}) saved by the best fixed judge-free design and added by the best evaluator on the same design, both as shares of uniform-sampling labels (retrieval: Qwen3-8B, $\epsilon = 0.02$; MT: best of 31 metric variants post hoc, pilot-fixed coefficient, $\epsilon = 0.01$).

(1 - ρ^2)(1 + 1/ P_{eff}) times that of the uncorrected one and

$$\text{HES} \approx \left[\rho^2 - \frac{1 - \rho^2}{P_{\text{eff}}} \right] \left(1 - \frac{P}{J_0} \right), \quad J_0 = P + L_0. \quad (2)$$

Without homoskedasticity the variance of $\hat{\lambda}$ takes a sandwich form and the tax term changes accordingly; the semi-synthetic evaluators below satisfy the assumption, the real ones need not. The proposition is bookkeeping of known results, an interpretive approximation rather than a theorem; J_{50} is a median-power budget, which under the normal

approximation scales with the expected variance. The first factor combines the PPI++ variance reduction ρ^2 (Angelopoulos et al., 2023b) with the classical inflation of a regression estimator whose coefficient is estimated (Cochran, 1977); the second is the share of the cost that remains after the pilot. The derivation is in Appendix A. Equation (2) implies that an evaluator with $\rho^2 < 1/(P_{\text{eff}} + 1)$ costs labels, about $\rho < 0.2$ for the 50-item MT pilot, where half the paired differences are zero.

Controlled variation. To test (2) over the whole range we built semi-synthetic evaluators, human utility plus Gaussian noise scaled to a target $\rho \in \{0.1, \dots, 0.9\}$, and ran them through the same audits (MT pilots 50 and 100, three Arena pairs; 126 cells). Predicted and realised HES correlate 0.975 (mean absolute error 0.035; Figure 4 in App. E); the tax-free ceiling does almost as well (0.974, 0.040) and the law over-predicts by 0.02, so the experiment supports the ρ^2 and dilution factors more than the tax term. The realised saving is 11% at $\rho = 0.5$ and 32% at $\rho = 0.8$. For the open evaluators on the two-system menus ($\rho \leq 0.36$) predicted and realised HES correlate only 0.20: realised savings (interquartile range -1% to +1%) are dominated by Monte-Carlo noise and exceed the ceiling by more than 0.03 in only 2-4% of cells.

The estimation tax on real metrics. Replacing the pilot-fitted coefficient by the population-

optimal one isolates the tax. Over the 317 metric variants it is 4.3 [3.5, 5.2] points on en→de and 2.5 [2.0, 3.1] on zh→en (metric-level bootstrap; H9). For the five metrics with the largest ρ^2 (0.11–0.15 on en→de, as $(\bar{\rho})^2$ or as PPSR’s mean ρ^2 , within the 0.09–0.15 PPSR reports), the realised saving over uniform sampling is between −4% and +6% (en→de) and −6% and +1% (zh→en), below the correlation-based saving in every cell (H10, under both definitions). With the oracle coefficient the same en→de metrics reach 6–12% on the decision-aware design: the asymptotic saving is not wrong, it is what an auditor who already knew λ would obtain.

Is the tax an artefact of freezing λ ? Mostly. Standard PPI++ fits λ on all labelled items. We reran the eight strongest metric variants per language pair, COMET-22 and the two GEMBA judges with the coefficient refitted on the pilot and every post-pilot label, and cross-fitted over two folds (App. B). Mean HES on the weighted design rises from +0.4% (pilot coefficient) to +6.8% with refitting and +1.6% with cross-fitting on en→de, against +6.7% for the oracle coefficient; on zh→en from −0.4% to +2.0% and +1.1%, against +1.3%; on the six chat pairs from +1.1% to +2.3% and +1.5%, against +2.1%. Its variance estimate ignores the coefficient’s estimation error, but over the 30 decisions and the six chat pairs the refitted upper bound covered the true difference at 0.90 (nominal 0.90), also at the budget nearest each design’s J_{50} , where it certifies (refit 0.902, cross-fit 0.903, pilot 0.903, human-only 0.904), and under boundary stress its largest type-I error was 0.05. Cross-fitting recovers less because each fold’s coefficient rests on half of the labels; a practitioner who needs no budget plan can refit, one who wants the variance estimate to account for the coefficient should cross-fit or freeze (the bound itself is calibrated by simulation, not proved). Refitting can exceed the oracle, which is the population least-squares coefficient, optimal for uniform sampling only. Across the 30 decisions (Figure 2b) refitted HES follows the tax-free ceiling $\rho^2(1 - P/J)$ (median deviation −0.4 points, correlation 0.42 over 1,158 decision–evaluator points), whereas with the pilot coefficient it falls a median 2.3 points below it; 94% of the ceilings are below 5%, because decision-level ρ is at most 0.43 and the pilot is a median 39% of the cost (range 15–66%). The tax is therefore the price of committing to the coefficient before the audit (to

plan the budget, say), not a property of the evaluator; refitting remains exploratory (the same labels set the coefficient and the inference). Even refitted, in the four locked cells the mean of the eleven (6.9, 6.7, 4.8, −0.3%) stays below the design (15.9, 9.0, 8.4, 8.5%).

Pilot size trades tax for fixed cost. Equation (2) predicts that a larger pilot lowers the tax $(1 - \rho^2)/P_{\text{eff}}$ but raises the fixed cost P . Both effects appear. Doubling the MT pilot to 100 segments shrinks the losses of the weak LLM judges (Mistral-7B, en→de, $\varepsilon = 0.01$: −7.1% → −0.9%) and halves the saving of a semi-synthetic $\rho = 0.9$ evaluator (32%→15%) because the pilot becomes most of the cost. A smaller pilot does not rescue the evaluators either: with pilots of 25 and 10 segments on the 30 two-system decisions (49 and 60 informative cells, now including en→de; pilot share 27% and 13% of the cost; the 31 metric variants), the design saves a median 4.3% and 4.4%, the mean evaluator adds 0.7% and 0.3% refitted (−2.7% and −7.7% with the pilot coefficient, whose tax grows as $1/P_{\text{eff}}$), and the design saves more in 86% and 90% of cells (App. C).

7 Chat: close pairs and presentation order

On the six close Arena pairs the pairwise judges reach (pilot-median) decision-level ρ of −0.01 to 0.29, the length heuristic 0.07–0.21. The best judge saves at least 5% with an interval above zero in two pairs (Figure 5a in App. F; H6 required three). Position bias explains part of the low ρ : one presentation order gives ρ of 0.19 (Qwen3-8B) and 0.12 (Mistral-7B), both orders averaged 0.25 and 0.16, turning Mistral’s saving from −1 to −2% into +0.5%. The bias itself is corrected by the human labels; the noise it adds lowers ρ .

8 Pre-registered outcomes and validity

Table 2 lists every pre-registered hypothesis. The design-dominance and estimation-tax hypotheses held; H2, H5 and H6 did not, and H4 held only in MT. H2 (value over uniform sampling inflated by ≥ 5 points) failed because the open judges’ value was near zero on both baselines; H5 (selection by pilot-predicted cost beats selection by pilot accuracy) did worse in both domains (excess cost 3.2% vs. 2.3% in MT, 2.8% vs. 1.7% in chat; near break-even, a 50-item pilot cannot resolve a few points).

Hypothesis	Outcome
H1 design saving > best open judge (MT)	586 4/4
H2 inflation ≥ 0.05 (MT)	0/4
H3 HES tracks $\rho >$ accuracy ^a	held ^d
H4 pilot predicts cost ($r \geq .8$)	MT; chat no
H5 pilot selector beats accuracy	no (both)
H6 chat HES $\geq 5\%$ in $\geq 3/6$	2/6
H8 design $\geq$ best of 31 variants ^b	4/4
H9 estimation tax > 0	both pairs
H10 realised HES < ρ^2 (top 5) ^c	all cells
H11 HES tracks $\rho >$ system r	both

Table 2: Pre-registered hypotheses (locks v0.8, v0.9, v1.1). ^aSpearman over two-system menus. ^bwith the locked pilot-fixed coefficient; with a refitted coefficient the post-hoc best of the eleven strong evaluators exceeds the design in 2/4 cells. ^clargely implied by the dilution factor $1 - P/J$. ^dby the locked rule; in MT both correlations are near zero (0.07 vs. -0.03), so we read it as inconclusive there. H7 and H2' concern frontier judges and are recorded in Appendix D.

H4 (pilot predicts relative design cost) held in MT (0.82), not in chat (0.37).

Validity. Over all locked cells the observed wrong-certificate rate is at most 0.070 (Clopper-Pearson upper limit 0.105). Under boundary stress (the runner-up as candidate, ε at 0.9 times its regret) the largest observed type-I error among 882 chat cells is 0.143 at nominal 0.10 (0.14 on the v0.9 set); no cell is significantly above α after Bonferroni correction (0.5% of cells at $p < 0.05$), and a Student- t bound changes the maximum to 0.140. These are Monte-Carlo checks of an asymptotic bound, not finite-sample guarantees.

9 Related work

Judges and metrics as system rankers. JuStRank (Gera et al., 2025) and Gao et al. (2025) show that instance-level accuracy does not determine system-ranking quality and that ranking degrades for close systems; Zhang et al. (2026) separate judgment quality from downstream utility. In MT, system-level correlation is driven by outliers (Mathur et al., 2020), close systems are better compared by pairwise accuracy (Kocmi et al., 2021; Deutsch et al., 2023), correlations on close pairs are much lower (Deutsch et al., 2022), and such comparisons are often underpowered (Card et al., 2020). We follow quality one level further, to the labels a close decision requires.

Prediction-powered evaluation. Control variates over human and metric scores, with modest

savings (Chaganty et al., 2018); PPI and PPI++ (Angelopoulos et al., 2023a,b; Gligorić et al., 2024), active inference (Zrnic and Candès, 2024a), Auto-Eval and prediction-powered ranking on Chatbot Arena (Boyeau et al., 2024; Chatzi et al., 2024), noisy-judge inference (Chen et al., 2026), PRECISE (Divekar, 2026), and the bound of Dörner et al. (2025) that an evaluator no better than the evaluated model saves at most half the labels. Closest to us, Gao et al. (2026) rank WMT metrics by the annotation they save under prediction-powered evaluation (PPSR, the mean squared correlation of paired differences) and verify the saving by resampling, with uniform labelling over all systems and the coefficient estimated on the evaluation labels. PPSR answers which evaluator saves most; we ask what the saving is worth once the decision is local, the design is credited first and the coefficient is fitted on a pilot. The two are complementary: PPSR's ρ^2 is the first factor of our cost law.

Annotation design. Choosing which items humans label is studied for model comparison (Sawade et al., 2012), active testing (Kossen et al., 2021), IR test collections (Sakai, 2016; Li and Kanoulas, 2017), and within PPI by cross-prediction (Zrnic and Candès, 2024b) and stratification (Fisch et al., 2024). Our judge-free designs are simple instances, credited before the evaluator.

10 Conclusion: three numbers to check before an audit

For a close system decision, an evaluator's value is set by the correlation of the differences the decision depends on, diluted by the pilot's fixed cost and by what a judge-free design already obtains; WMT22 metrics and open LLM judges then save a few percent of human labels on average, the strongest somewhat more, and the typical one less than organising the annotation itself. The results reduce to a rule a team can apply before paying for an evaluator: (i) its decision-level ρ on the systems at stake, from a prior campaign or the pilot, not its system-level agreement; (ii) the pilot's share of the cost, P/J , which a human-only pilot prediction supplies; (iii) what the judge-free design already saves, known in advance. The increment is at most $\rho^2(1 - P/J)$, less $(1 - \rho^2)/P_{\text{eff}}$ of the post-pilot share if the coefficient is frozen: for the WMT22 metrics ($\rho \leq 0.43$, $P/J \approx 0.4$) at most 11%, typically 2–5%, against design savings of 5–16%.

Limitations

The menus are built from full human labels to be deliberately hard; easier decisions leave less to save for any method. The certificate uses a normal bound whose calibration we checked by simulation, not proved. MT savings are computed on the MQM-rated subsets of WMT22 (two language pairs), chat on six pairs from a 2023 Arena release, retrieval on re-analysed audits of an earlier study with two judges. The cost law is a first-order approximation; it describes controlled variation of ρ and bounds, but does not predict cell by cell, the savings of real evaluators. The LLM judges are open 7–8B models; frontier LLM judges are not evaluated here, and their decision-level ρ may be higher. The evaluator enters only as a control variate on the audit; its other uses (choosing the candidate, building the menu, stratifying the sample) are outside the decision value we measure. HES depends on the pilot size, and larger pilots lower the estimation tax while raising the fixed cost (§6). The results should not be read as showing that automatic evaluators are useless: they concern close decisions, small human pilots and evaluators with decision-level $\rho \leq 0.43$; by the cost law, an evaluator with $\rho \approx 0.8$ would save about a third of the labels with our pilots, and more with a smaller pilot share, but against single WMT22 ratings label noise caps ρ near 0.6 (App. C), so such savings also require more reliable human labels. Uncertainty is Monte-Carlo over simulated audits of one fixed test set per language pair; items and raters are not resampled. We report J_{50} ; at 80% power savings differ and are often censored by the budget grid (Appendix B). The estimand treats identical strings as one quality; sharing one random rating per string instead leaves the design savings essentially unchanged (App. C), but under an as-rated estimand, in which identical strings may differ in quality, their rating noise would be variance that deduplication does not remove. The retrieval evidence rests on a companion study under review.

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A Derivation of the cost approximation

Write D for the paired difference of an informative item, $\sigma^2 = \text{Var}(D)$, and $\hat{D}$ for the evaluator’s prediction with $\text{Corr}(D, \hat{D}) = \rho$. For a

fixed coefficient λ the residual $R = D - \lambda\hat{D}$ has variance $\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2$, minimised at $\lambda^* = \rho\sigma/\hat{\sigma}$ to $\sigma^2(1 - \rho^2)$ (Angelopoulos et al., 2023b). With $\hat{\lambda}$ the least-squares coefficient from a pilot of effective informative size P_{eff} , independent of the post-pilot sample, and a homoskedastic residual ($E[R^2 | \hat{D}] = \sigma^2(1 - \rho^2)$), $E[(\hat{\lambda} - \lambda^*)^2] \approx \sigma^2(1 - \rho^2)/(P_{\text{eff}}\hat{\sigma}^2)$, so the expected residual variance is $\sigma^2(1 - \rho^2)(1 + 1/P_{\text{eff}})$ (Cochran, 1977, §7). Under the normal fixed-budget approximation the post-pilot labels needed to bring $z\sqrt{v/L}$ below the slack scale with the variance factor, so $L \approx L_0(1 - \rho^2)(1 + 1/P_{\text{eff}})$ and $1 - L/L_0 \approx \rho^2 - (1 - \rho^2)/P_{\text{eff}}$ to first order in $1/P_{\text{eff}}$. With the pilot cost P common to both arms, $\text{HES} = 1 - (P + L)/(P + L_0) = (1 - L/L_0)(1 - P/J_0)$. Clipping λ at zero bounds the loss of an evaluator with $\rho \leq 0$ by the variance of $\hat{\lambda}$ only; zero-difference items reduce P_{eff} below P , which is why MT, with about half its paired differences zero, pays more tax. In general, $\text{Var}(\hat{\lambda}) \approx E[(\hat{D} - E\hat{D})^2 R^2]/(P_{\text{eff}} \text{Var}(\hat{D})^2)$ (sandwich form), which reduces to the expression above under homoskedasticity; heavy-tailed or heteroskedastic residuals, as with sparse MQM differences, raise the tax.

B Robustness of the estimation tax

Mean HES over the evaluators of each set, over their own judge-free base, by how the coefficient is set (J_{50} ; MT: eight strongest metric variants per language pair, COMET-22, Qwen3-8B, Mistral-7B; chat: Qwen3-8B, Mistral-7B, longer). Pilot: fitted on the pilot and frozen (the paper’s estimator); refit: refitted by unweighted least squares on the pilot and every post-pilot label; xfit: cross-fitted over two folds; HT: the same with $1/\pi$ -weighted least squares; oracle: population-optimal. COMET-22 enters twice in MT, as the WMT22 submission (COMET-22-refA) and as our own run; their segment scores correlate only 0.78–0.79, and dropping ours changes the MT means by at most 0.3 points.

Cell	base	pilot	refit	xfit	refit _{HT}	xfit _{HT}	oracle
MT en→de	weighted	0.4	6.8	1.6	10.0	−1.3	6.7
MT en→de	dedup	0.8	7.2	1.7	12.4	−1.3	–
MT zh→en	weighted	−0.4	2.0	1.1	2.0	1.1	1.3
MT zh→en	dedup	−1.8	2.3	1.9	2.5	2.3	–
Chat (6 pairs)	uniform	1.1	2.3	1.5	2.3	0.9	2.1

Values in %, averaged over ε . With J_{80} in place of J_{50} the MT values are 0.1–1.2 (pilot), 2.2–4.9 (refit), 1.5–2.8 (xfit), 2.1–4.4 (refit_{HT}), 0.7–2.9 (xfit_{HT}) and 1.9–2.5 (oracle); J_{80} is often near the

largest budget, so these are less reliable. Under boundary stress the largest type-I errors were 0.04 (pilot), 0.047 (refit), 0.043 (xfit), 0.07 (refit_{HT}), 0.05 (xfit_{HT}) and 0.06 (human-only); no cell was significantly above α . Evaluator-driven allocation (sampling by the pilot-estimated residual variance, as in active inference) reduced J_{50} relative to the weighted design with the same evaluator by -1.6% to $+1.7\%$ on average over the 31 metric variants (individual cells -11% to $+19\%$).

C Sensitivity of the decision analysis and a human noise ceiling

All exploratory and post-lock (code/123_decision_sensitivity.py, code/124_human_ceiling.py).

Informativeness threshold. The post-pilot share $1 - P/J$ of the 60 MT cells takes no value between 0.21 and 0.47, so every threshold in that range selects the same 33 cells. With a threshold of 0.2 (34 cells), 0.5 (30) or 0.6 (23; one en $\rightarrow$ de cell) the design’s median saving is 5.4, 5.8 and 5.2%, the mean evaluator’s refitted HES 1.5, 1.5 and 1.7%, and the design beats the mean evaluator in 94/82/91, 93/83/90 and 91/78/87% of cells (pilot/refit/cross-fit).

Evaluator set. Restricting the mean to the ten evaluators per language pair with the highest system-level Pearson, chosen without reference to any decision, raises the mean evaluator’s median HES on the 33 cells to 0.6, 2.8 and 1.8% (pilot, refit, cross-fit); the design still saves more in 91, 76 and 79% of cells.

One rating per identical string. WMT22 rated identical strings separately, and our estimand averages their ratings. To check that the deduplication saving does not depend on this averaging, each group of identical strings instead shares one of its ratings, drawn at random (`-ident pick`). In the four locked cells the best fixed design then saves 17.0, 8.5, 9.2 and 12.2% (means over three draws of the shared rating; ranges 15.5–18.4, 7.9–8.8, 7.9–10.0 and 11.0–13.9) instead of 15.9, 9.0, 8.4 and 8.5%. Over the 30 decisions (again 33 informative cells) the design’s median saving is 5.2% instead of 5.5%, the mean evaluator’s median HES -0.6 , 1.4 and 0.1% (pilot, refit, cross-fit) instead of -0.1 , 1.6 and 0.4%, and the design beats it in 91, 85 and 91% of the cells instead of 94, 82 and 91%. The

deduplication saving therefore does not rest on the averaging.

Dissimilarity bins. The weighted design bins the positive chrF dissimilarities into three groups (below the median, third and fourth quartile; identical outputs have weight zero). With four quartile bins instead, the design’s J_{50} in the four locked cells changes by at most 0.8% (weighted: 390 vs. 389, 238 vs. 238, 2,743 vs. 2,767 and 1,103 vs. 1,095 labels), the best fixed design’s saving by at most 0.1 points, and the best metric’s increment by at most 1.2 points; over the 30 two-system decisions (the same 33 informative cells, 31 metric variants) the design’s median saving is 5.5% under both binnings and the mean evaluator’s HES $-0.5 / 1.3 / 0.2\%$ (pilot / refit / cross-fit) against $-0.1 / 1.6 / 0.3\%$ with three bins (supplementary results/VARIANTS.md).

Monte-Carlo precision. With 2,000 audits per cell (31 metric variants) the informative cells are 36 rather than 33 (en $\rightarrow$ de decisions 13 and 24 at $\varepsilon = 0.01$ and zh $\rightarrow$ en 05 at $\varepsilon = 0.02$ cross the 0.3 threshold, with post-pilot shares 0.37–0.42); the design’s median saving is 4.6% (300 audits: 5.5%), the mean evaluator’s 0.2 / 2.3 / 0.9% ($-0.1 / 1.6 / 0.4\%$), and for the ten strongest evaluators 1.4 / 3.9 / 2.2% (0.6 / 2.8 / 1.8%); the design saves more in 100 / 89 / 100% of cells (89 / 61 / 78% against the ten strongest). The one-sided bounds of §5 are in supplementary results/EXCLUSION_n2000.md; the 300-audit runs remain the paper’s primary analysis because they include the three LLM judges.

Pilot size. The 30 two-system decisions re-run with pilots of 25 and 10 segments (the 31 metric variants; the same menus, ε and coefficient rules). Informative cells: 49 and 60 (pilot 50: 33); pilot share of the cost, median: 27% and 13% (35%); design saving, median: 4.3% and 4.4% (5.5%); mean evaluator HES, median over cells, pilot / refit / cross-fit: $-2.7 / 0.7 / 0.0\%$ and $-7.7 / 0.3 / 0.0\%$ (pilot 50: $-0.1 / 1.6 / 0.4\%$); for the ten evaluators with the highest system-level correlation $-2.4 / 2.2 / 0.0\%$ and $-6.8 / 0.5 / 0.0\%$; the design saves more than the mean evaluator in 98 / 86 / 92% and 100 / 90 / 98% of cells. The frozen-coefficient tax grows as the pilot shrinks, as (2) predicts, and refitting leaves the increment below 1%.

Evaluators on the weighted vs. the dedup base. Lock v1.1 applied every metric on the weighted

design and compared it with the cheapest judge-free design, which on zh→en is dedup; Table 1 now applies the metrics on that same base. On zh→en at $\varepsilon = 0.01$ the best metric adds 7.3% [−6, 19] on dedup against −0.9% on weighted; at $\varepsilon = 0.02$, 1.3% against 0.8%. The H8 verdict is unchanged either way, and the locked numbers are kept in results/MTME_VERDICTS.md.

Human noise ceiling. WMT22 re-annotated en→de with three raters per item. For each of the 15 pairs of the top six systems, the paired differences of two distinct raters (drawn at random per item, 200 repetitions) correlate 0.31–0.41 (median 0.35; the same without identical outputs). Since the audits’ labels are single ratings, an evaluator that knew the true difference would reach at most $\rho \approx \sqrt{0.35} = 0.59$ (0.55–0.64 per pair) against them, and $\rho^2 \approx 0.35$ caps its tax-free saving at about a third of the post-pilot labels. The best en→de metrics (ρ 0.30–0.38) reach one half to two thirds of that ceiling, and their ρ^2 a quarter to two fifths of it, so both label noise and the evaluators limit the saving.

D Pre-registration record

Lock v0.8 (MT, chat; hypotheses H1–H5 and validity) was committed before any LLM or COMET judgment was joined with a human label; the human-only MT designs and chrF had been examined in development and are reported as such. Two deviations followed (a type fix and an accounting fix charging one Arena vote per battle). Lock v0.9 replaced the v0.8 chat decision, whose pairs had win-rate gaps of 0.10–0.56 and were certified by every design right after the pilot, by the six closest pairs; no judge output existed for these battles. Lock v1.1 fixed the 31-metric test (H8–H11) before any of its savings were computed; H10 as locked uses $(\bar{\rho})^2$ and is reported with PPSR’s mean of squared correlations as a post-lock robustness check, and H9’s interval is a metric-level bootstrap. Lock v1.0 (frontier judges) is committed with its exported inputs and scoring rule and has not been run. Analyses not listed in a lock (three-level anatomy, oracle- λ tax for open judges, presentation order, semi-synthetic ρ) are exploratory.

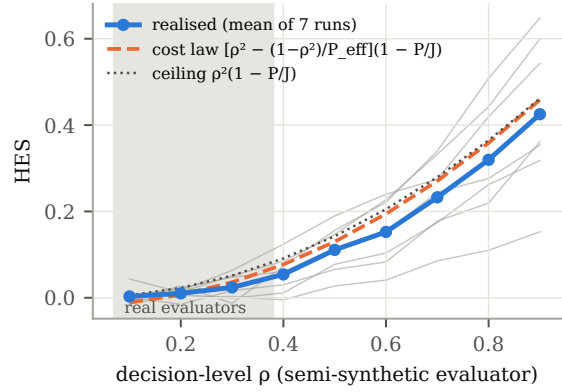


Figure 4: Realised HES of semi-synthetic evaluators with a set decision-level ρ (grey: individual runs) against the cost law (2) and its oracle- λ ceiling. Shaded: the range of the real evaluators.

E Controlled variation of ρ

F Chat audits

Figure 5 shows the six close LMArena pairs of lock v0.9 (§7).

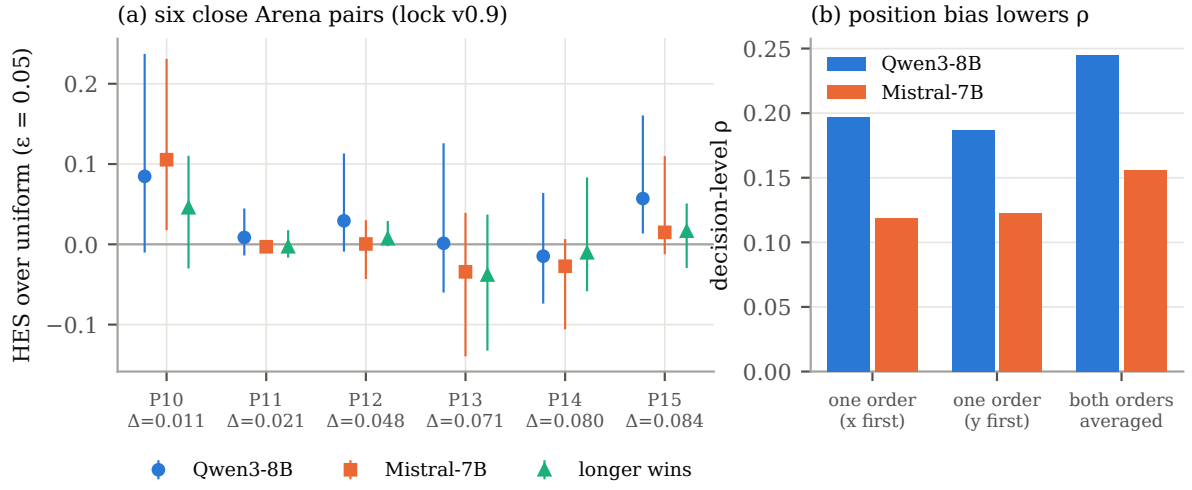


Figure 5: (a) HES over uniform sampling of three evaluators on six close LMArena pairs (Δ : human win-rate gap; pilot 50, $\varepsilon = 0.05$; 95% intervals). (b) Decision-level ρ with one presentation order and with both orders averaged.